Asymetryczne algorytmy szyfrowania
Domaradzki Mikołaj

1.2

RSA 512 bit

Prywatny:

```
-----BEGIN PRIVATE KEY-----
MIIBVAIBADANBgkqhkiG9w0BAQEFAASCAT4wggE6AgEAAkEAsEqh5nDcC9x43cQ/
IN7sRvM9axXfxhGn6mAxT6NiVsg8w/RpmUmJstUhk5y/15Bdfevo60mPAAu2Bswo
pLCRawIDAQABAkBVBtoJZButmNuu0m1ENGCvY7gWwwkX5gIf998wsvPGOKYGMrK5
P6dPOAlfjQsTg1HDk1Yc2fbS45cuhoH6qzdJAiEA5kzxhsuMEzHuj3aL53tNX2gM
tVSC+4NJSJH7Hvh9UBcCIQDD9syErZFmaXONeeaaoCkALxVH4ikBYiByo4JzEjJp
zQIgcTZX8TS0umbsaFgVdIj3wzPqM2XFMhEbAHLK6TGsoKcCIQClwTnY+B9Gques
wXGHVPWZTDoJ5YidvOyJSzjqMHgn+QIgdeUfEEdZMlvZAxiECQGyVM2FC6/T8you
mbgAT8kC8No=
-----END PRIVATE KEY-----
```

Publiczny:

```
-----BEGIN PUBLIC KEY-----
MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBALBKoeZw3AvceN3EPyDe7EbzPWsV38YR
p+pgMU+jYlbIPMP0aZlJibLVIZOcv9eQXX3r6OtJjwALtgbMKKSwkWsCAwEAAQ==
-----END PUBLIC KEY-----
```

RSA 1024 bit

Prywatny:

```
-----BEGIN PRIVATE KEY-----
MIICdQIBADANBgkqhkiG9w0BAQEFAASCAl8wggJbAgEAAoGBAJ6Bwp2rC7U/zmEh
9Xj4KPnRuJJjmAnmilsIH2Pz2iy8uDGefrsKnnPC1NM2/vbqCs/RoPYoGPtiPq8k
83hUxrDaDbft5DC+HXcjivGFSjmPjp1abhtayt+zqblEyOm+Kiu21VmKbifZl98T
f1Ftw1ED8VeGY1dlsoV+mBR+W5UjAgMBAAECgYBYKsl8NjZu/Szo7qDCRxdTzooi
M2/QogH3nM8JJA0JsXercgQAOmnk7ky/p+muooSdEHe+NJOg0QcZGS6skktqF+cW
JNIHApXMQ0zTJ4vyNlpG4LNfFiz2curkzKvqJjWmUsgu7fygCaVy20I78Pc/Prl6
TG9u/gNQW2xD3aJOQQJBAMm0ILOLDGeUaJT7egW68IGKmuepeYDVvQAu/TrfZeAU
YsNSyZfQcEXOHYw4vuDG5bKT0SrmwfqZ1XW8bbvj2LECQQDJLNe82qORehtfUbF5
TUoFkjUhVTbGVhIHyGC4dapWD0rZ7inv3cpo7RNroJQ948TB2PIrDAe/1A+JPWkI
OoATAkAvVDcAKjaE74wCuWeOhKj+nJlwlpHOK1kn+96ru9PZ2q66CmkHWH+fwZep
WX9gmws29fx0xqTpusMd11MgbXxhAkA4sK5KxcH6KNK11auRZzpIv57BOeYple+W
ydyXnnDVt5F8QCUJNxLm5ofmeGXN2vWDOsNjZgx4/iUyo1+JthKJAkBpj9Lzondy
QbctGSlsNnjOAU0w9/810vyKgf8ykj4frhA78pGS1eLLxeoGs86xb251EdT9Ax6X
IlR06PUj2TMu
-----END PRIVATE KEY-----
```

Publiczny:

```
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQCegcKdqwu1P85hIfV4+Cj50biS
Y5gJ5opbCB9j89osvLgxnn67Cp5zwtTTNv726grP0aD2KBj7Yj6vJPN4VMaw2g23
7eQwvh13I4rxhUo5j46dWm4bWsrfs6m5RMjpviorttVZim4n2ZffE39RbcNRA/FX
hmNXZbKFfpgUfluVIwIDAQAB
-----END PUBLIC KEY-----
```

RSA 2048 bit

Prywatny:

```
-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBKcwggSjAgEAAoIBAQCHH7EA1vDFMtWD
fzG4l/hP5iID0ShCUwILCdC9rqntGYCqCb0PvXhpHQ2GbcOPT8K8xoGMVvjVQbuN
Vl1QDewE8/52ooMZeJuYzLiL3D5DWU0wmQdF3VHJOyyYvoJCV1LyrniP0UMCk1mZ
E6iQXdBOyHGQYfvhMfLkPIzUfj5WHef/de2PrzmO2GdtPxAW9TQ+8y5M/o/4aFcX
5pNqU4cyjzowYk5K2lXwBKhUrwqasB+gKjREOyJ5hBhst7Ebi6oZXDoVHT1LLWic
KjJg5nb+wH+BEHCJLlKwhDDtXnKWj57+2H6OwP4gp16IO1BBF+iosThVkXnzNKJm
fGJD5AYvAgMBAAECggEAJPUpD4a6geU+euHeMpP2SbwzD4I0lbdodqZvnoHYXk48
X8WcHWMAPRfCLEN3WopBRCqzIjRjiMiayLSXfx7fsBcgQUh/vQa84BUWYMKTkQ4q
y6D/WT+IcBasQvbQX97MhC3uMb/VqS2uMu0qjJnsMLAoR4pVPynPOlfJWRFScXEE
0pp0+dqM7lbOZwdlY1yxLWPTgvE8UbHAVZ3QKOPZhS5z0eDC6QwMDfEtQq387E3h
13cn7glTnY7+LEmBtZuzN0Zig8+KhPYrQ6ZZJ4tR4Syy9pFxLoifIVy9Iru22uVv
z1zAGeiktESDDCfJ5Gn6jcO9r8+Uk37sCIENxWYcQQKBgQC9U0OHHsO0MsJkMIPx
+S6b6QbeCUObm0z3LctR3x/5QybgjUIZkwecRPsyc8YBHWgKnzqpPires8OHhV+U
disuceqsisNsbH2imZe5GX3sbYuVXBq1aV5Svgw+Ka2jzMNAED6jZkxVqCR65H60
Z2/Bm369n+daFMchO2L+NJbzYQKBgQC2tdySrnPULqGI2FyfrwFp7uFgCO+fMD88
VVwywIsjb5Ym0W7jEnhKulCqkN83h42L2esGPmMTPqq7rGh2608E2+fzkPoNXC9D
ZFYM8HRU+aBeTuKgJXwzDCJ6tJjVqdE+vm9cf4yg3B+OJ2cA4FQIm0gqtbQdvWMS
Q7+KYq3zjwKBgGTWZk8fiaj+qB7o4D4PNE4CTZARaxsEGnhEwdg3eTYwGBLd6ejp
MNKAwzJaC4ooeXO+32WpIPhA0tHprcDLN7CjKBUCS5gCKOBj4jXKaBF3PzMsPWWG
W6DCbI7nb/6gXxZOEK7DG8E7r0JtVTcNKK77/QNOmdCW5IDEurMem4/BAoGAXO8+
JaH0qq2ZXpjm7BZBOtfV7Vo+QBNzZsXoYqxLWZftoQsQQxHm++OvGkMcNwuM9ODA
Sem7zVMn1tQIQszH4anVXq+UcO094SKNHoHEOmDKsI2hemH2vaRjeVSZFL0PAMIu
tLbwGfY9WroyC2EPfV02pPrNhf8KAIVTTRmKb4MCgYEAh4RC4Qmp8nMpNTrDtlhq
KI8TNtXeGLJLAhsY0Su+bL0VoYkJslPKVglB0L4dKhS7XV+OdF8J6BSO3y+XhC71
v5Q14rm6EsSq5j1Ciipu15A8/z3Mo9XlcpwLdyl3t6lEPTBkbCgkzIfY+9W7spaR
h+5AdMR9MUAdpNcAzZg74fc=
-----END PRIVATE KEY-----
```

Publiczny:

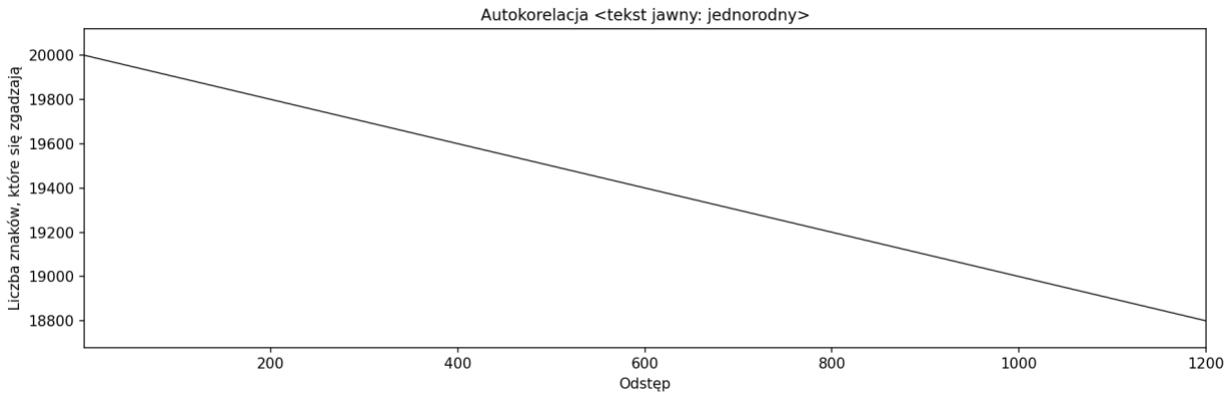
```
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAhx+xANbwxTLVg38xuJf4
T+YiA9EoQlMCCwnQva6p7RmAqgm9D714aR0Nhm3Dj0/CvMaBjFb41UG7jVZdUA3s
BPP+dqKDGXibmMy4i9w+Q1lNMJkHRd1RyTssmL6CQldS8q54j9FDApNZmROokF3Q
TshxkGH74THy5DyM1H4+Vh3n/3Xtj685jthnbT8QFvU0PvMuTP6P+GhXF+aTalOH
Mo86MGJOStpV8ASoVK8KmrAfoCo0RDsieYQYbLexG4uqGVw6FR09Sy1onCoyYOZ2
/sB/gRBwiS5SsIQw7V5ylo+e/th+jsD+IKdeiDtQQRfoqLE4VZF58zSiZnxiQ+QG
LwIDAQAB
-----END PUBLIC KEY-----
```

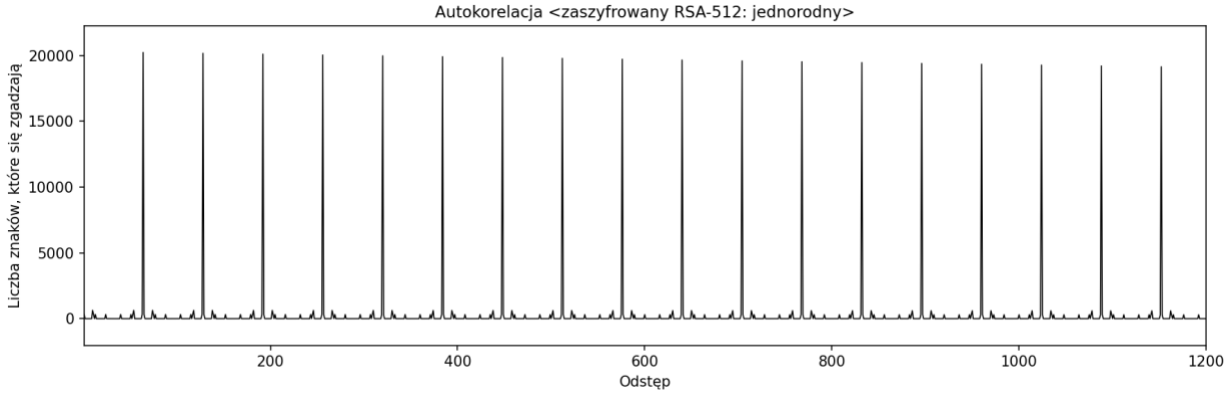
1.3

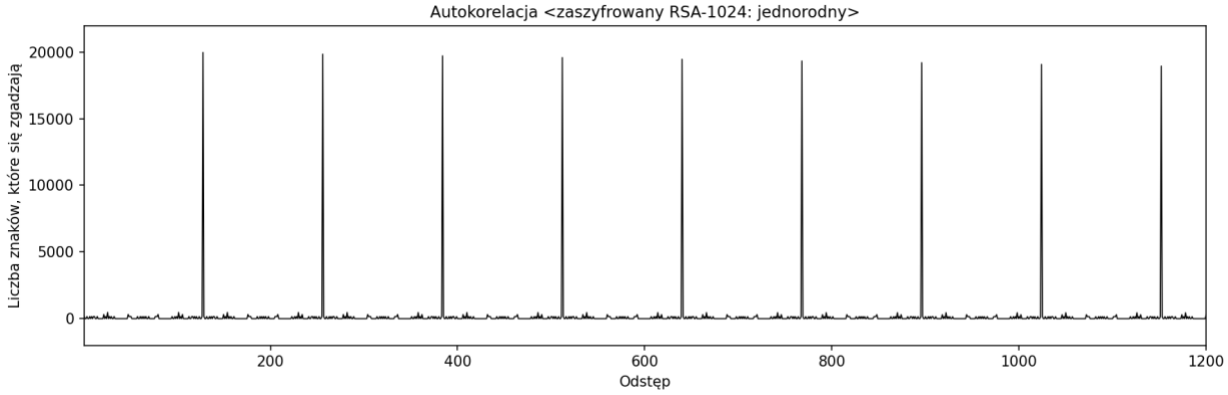
Entropia

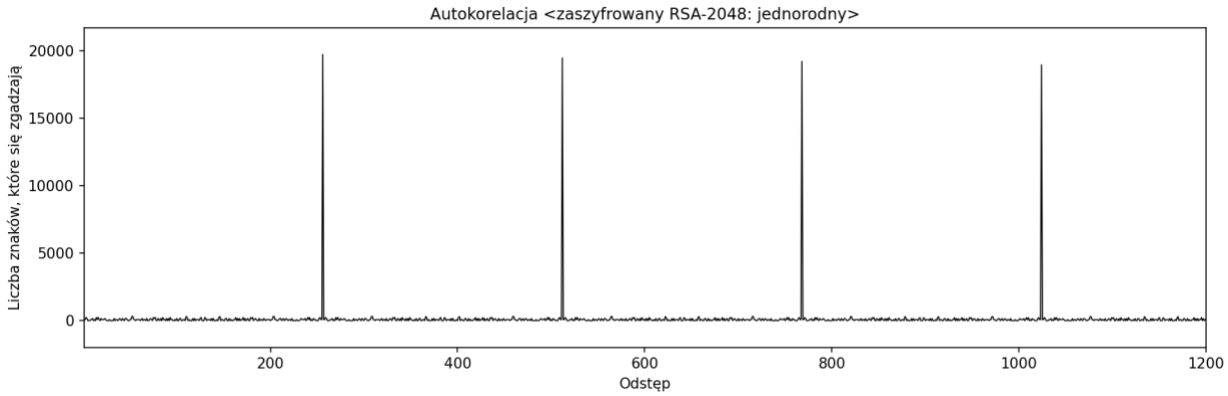
	TJ	RSA 512	RSA1024	RSA 2048
Jednorodny	0.0000	5.8345	6.6947	7.2233
Średnio zróżnicowany	4.1464	7.9909	7.9910	7.8025
Wysoce zróżnicowany	6.5658	7.9916	7.9912	7.9914

Tekst jednorodny

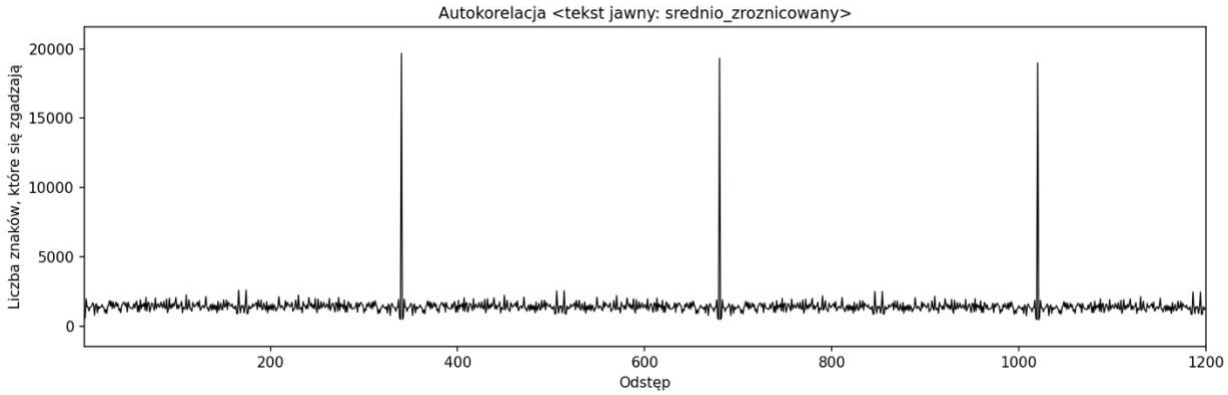

Autokorelacja <tekst jawny: jednorodny>
Liczba znaków, które się zgadzają
20000
19800
19600
19400
19200
19000
18800
200
400
600
800
1000
1200
Odstęp

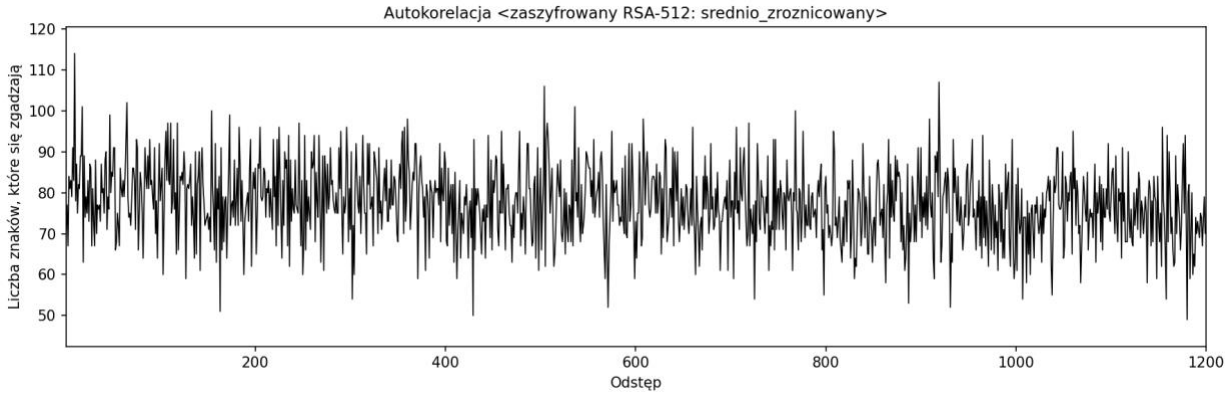

Autokorelacja <zaszyfrowany RSA-512: jednorodny>
Liczba znaków, które się zgadzają
20000
15000
10000
5000
0
200
400
600
800
1000
1200
Odstęp

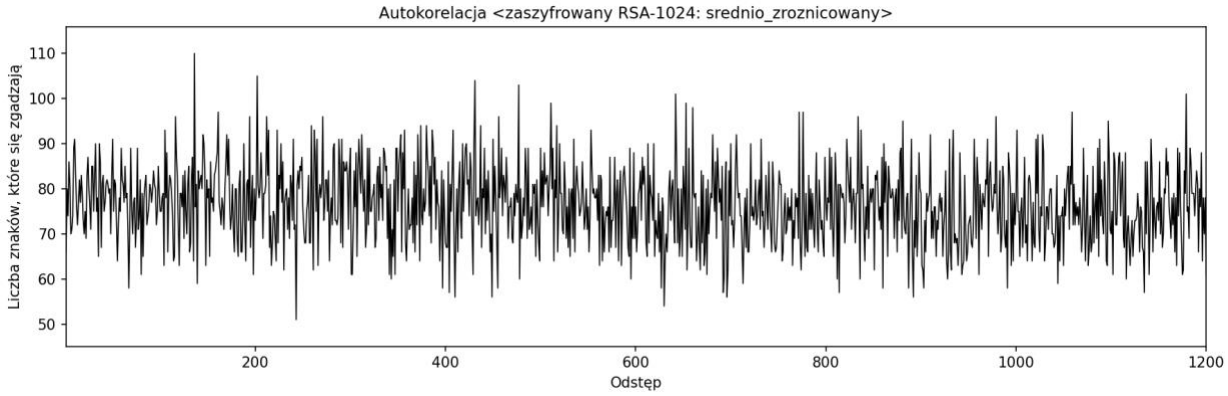

Autokorelacja <zaszyfrowany RSA-1024: jednorodny>
Liczba znaków, które się zgadzają
20000
15000
10000
5000
0
200
400
600
800
1000
1200
Odstęp

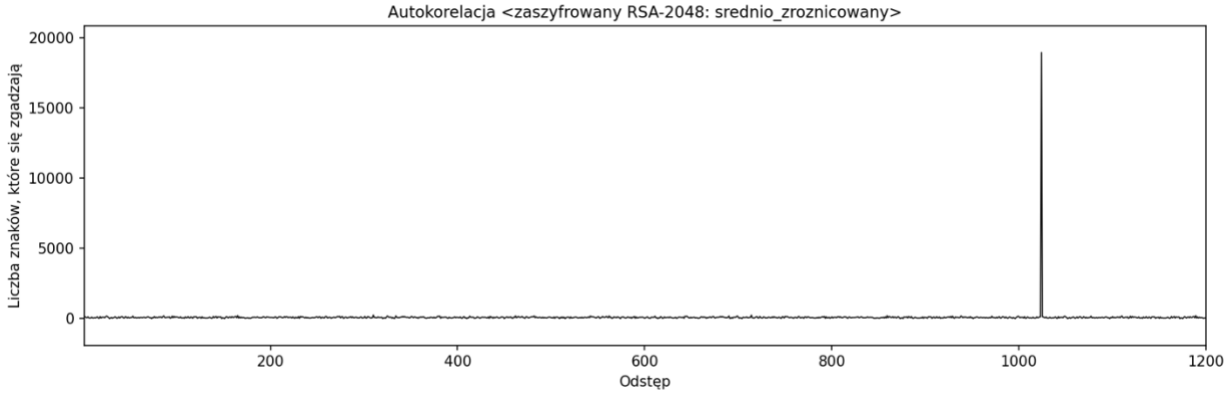

Autokorelacja <zaszyfrowany RSA-2048: jednorodny>
Liczba znaków, które się zgadzają
20000
15000
10000
5000
0
200
400
600
800
1000
1200
Odstęp

Tekst średnio zróżnicowany

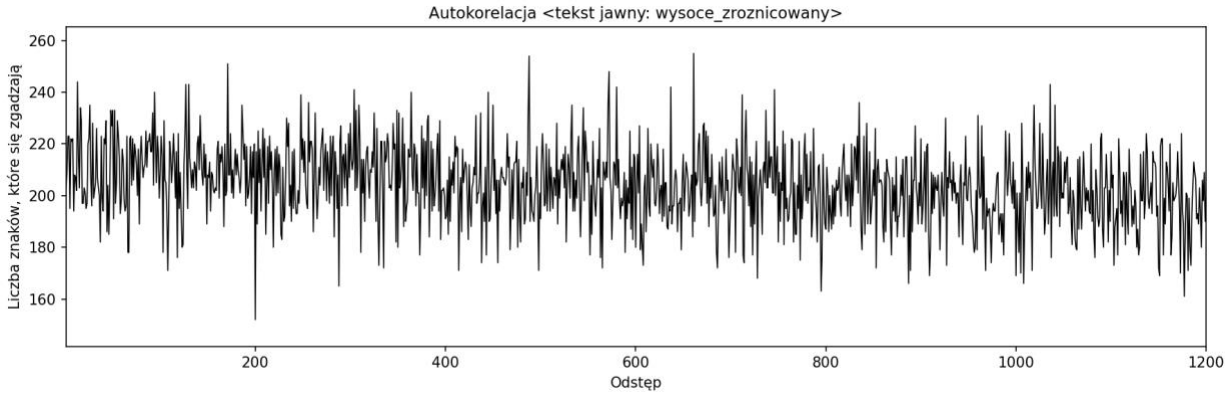


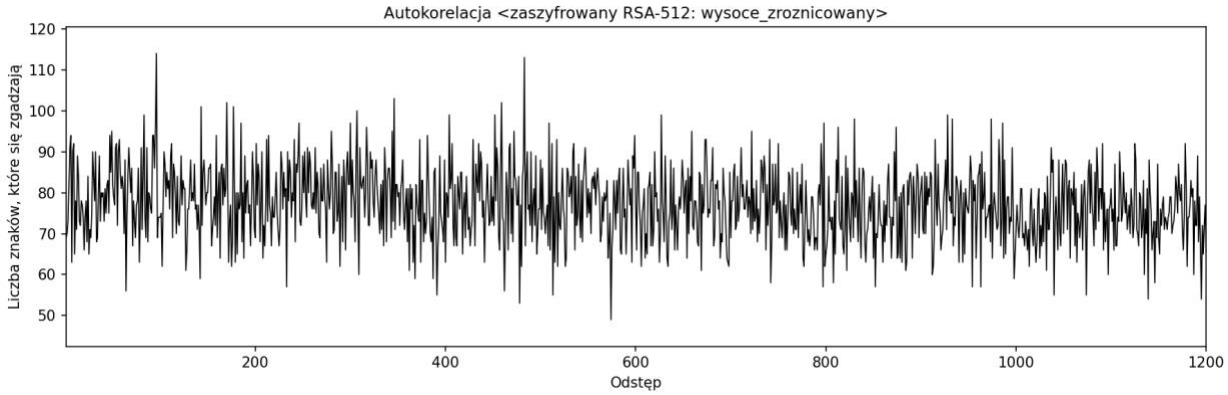


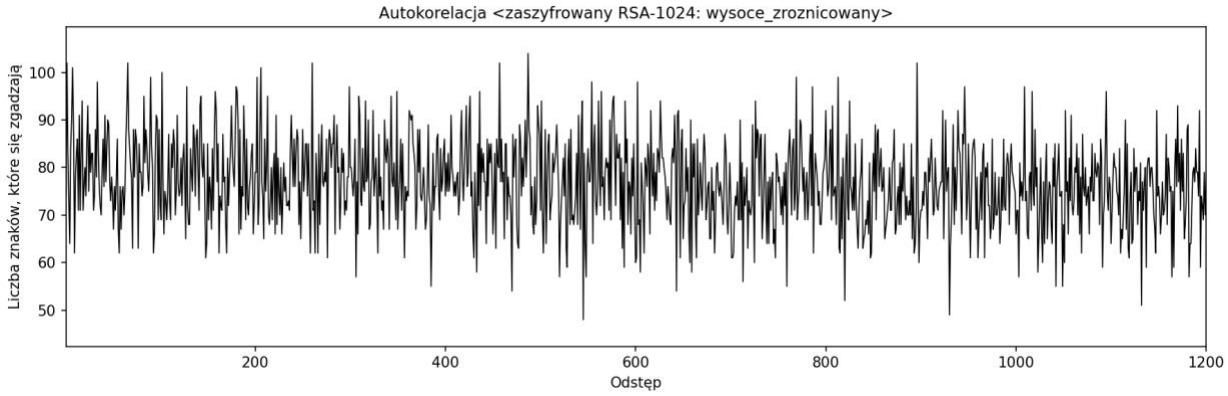


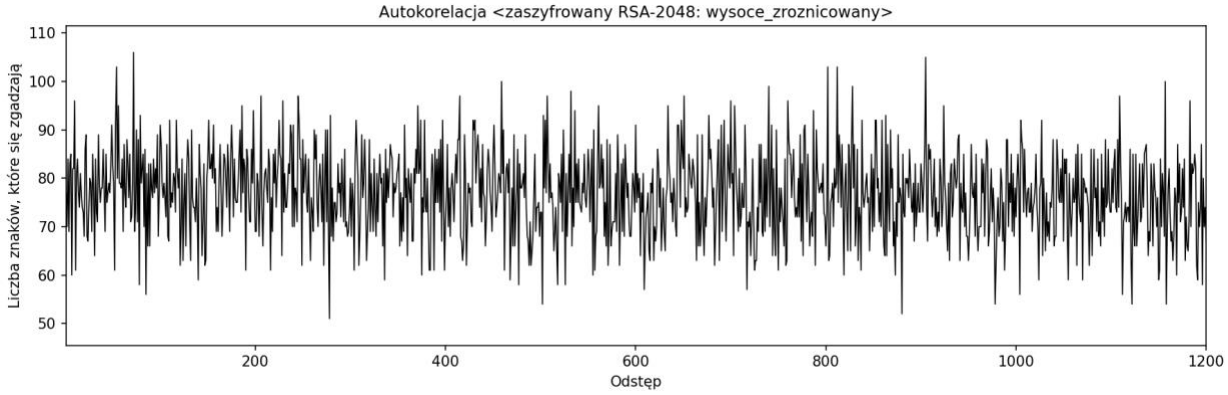


Tekst wysoce zróżnicowany









1.4

Wyniki w [s]

	RSA 512	RSA 512	RSA 1024	RSA 1024	RSA 2048	RSA 2048
	Szyfr	Deszyfr	Szyfr	Deszyfr	Szyfr	Deszyfr
1 MB	0.81	1.84	0.66	1.74	0.64	2.93
2 MB	1.66	3.75	1.34	3.43	1.29	5.87
5 MB	4.17	9.41	3.36	8.66	3.28	15.17

1.5

Wyniki w [s]

	AES 128	AES 128	AES 192	AES 192	AES 256	AES 256
	Szyfr	Deszyfr	Szyfr	Deszyfr	Szyfr	Deszyfr
1 MB	0.0040	0.0034	0.0040	0.0037	0.0043	0.0042
2 MB	0.0070	0.0067	0.0078	0.0075	0.0086	0.0084
5 MB	0.0177	0.0170	0.0188	0.0193	0.0213	0.0211

1.6

Tekst średnio zróżnicowany

RSA 512

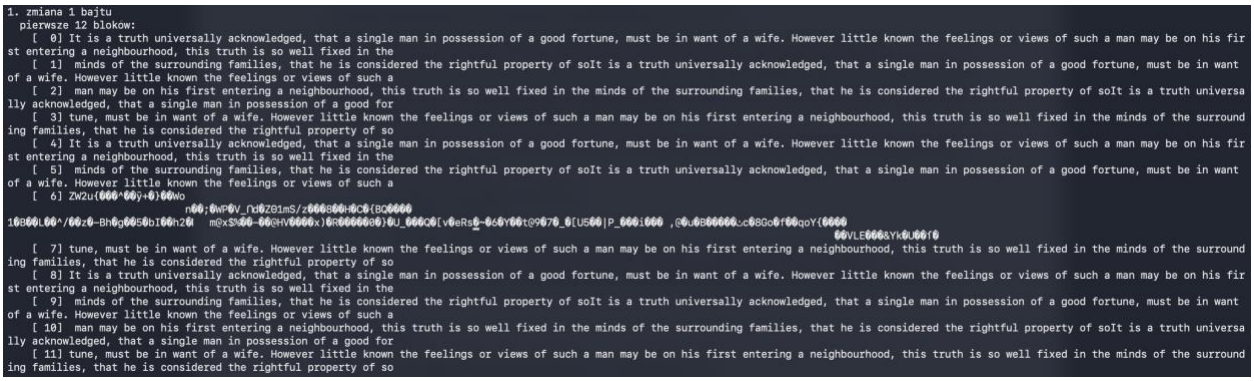
```
1. zmiana 1 bajtu
  pierwsze 12 bloków:
    [  0] It is a truth universally acknowledged, that a single man in po
    [  1] ssession of a good fortune, must be in want of a wife. However
    [  2] little known the feelings or views of such a man may be on his
    [  3] first entering a neighbourhood, this truth is so well fixed in
    [  4] the minds of the surrounding families, that he is considered th
    [  5] e rightful property of soIt is a truth universally acknowledged
    [  6] ??d?:>;>&Uv¡?F3X5??}?????????K_)L???%?JS?
#??[?????
    [  7]  want of a wife. However little known the feelings or views of
    [  8] such a man may be on his first entering a neighbourhood, this t
    [  9] ruth is so well fixed in the minds of the surrounding families,
    [ 10]  that he is considered the rightful property of soIt is a truth
    [ 11]  universally acknowledged, that a single man in possession of a
```

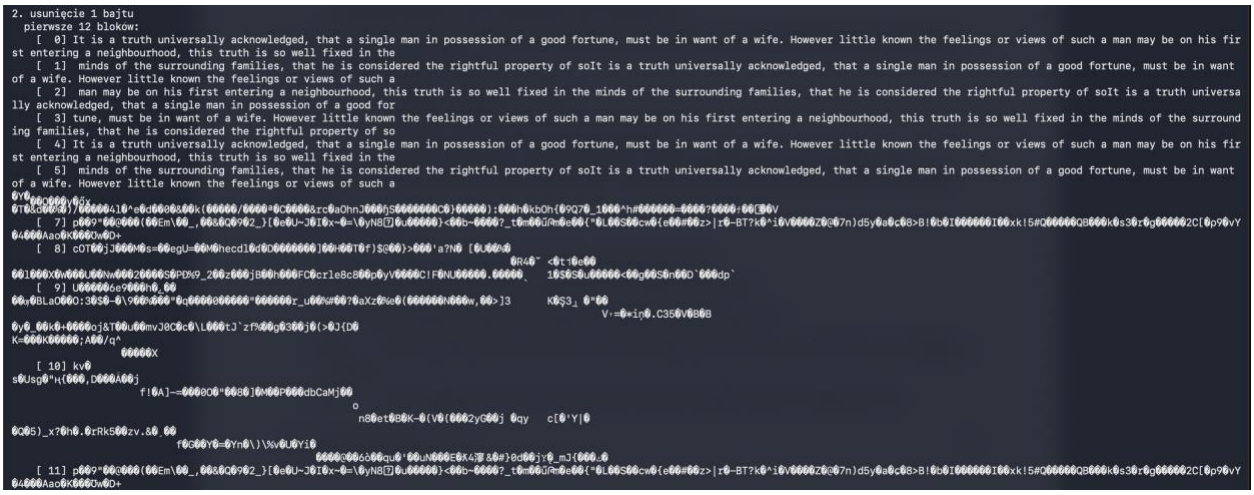
```
2. usunięcie 1 bajtu
  pierwsze 12 bloków:
    [  0] It is a truth universally acknowledged, that a single man in po
    [  1] ssession of a good fortune, must be in want of a wife. However
    [  2] little known the feelings or views of such a man may be on his
    [  3] first entering a neighbourhood, this truth is so well fixed in
    [  4] the minds of the surrounding families, that he is considered th
    [  5] e rightful property of soIt is a truth universally acknowledged
X??: K?K{?5cQ?ui???bŧ B????S?T?s??       ?h??
    [  7] 6h??h#?₥f?p[l?20?????'q7L?>[A~??)??i#?)???p??Z?3???T6?
    [  8] ?O?9RE#g?s??d!????????23???O-y·c???J%?)???,xv???R(???R
    [  9] ?v?P??t???kf.0N?i??l·???J>?M?獸?jB?J??{?O#¿???P?
    [ 10] ???B?7,I?
                  Q]k?]&2PM$???v?ˌ?r?L(? ?wK??η
                                          ??????P???6%C??
    [ 11] ?r?²
              j?OD??l?5}~i????S!???DřSD???o??f}x?J?????ni(9H??T?
```

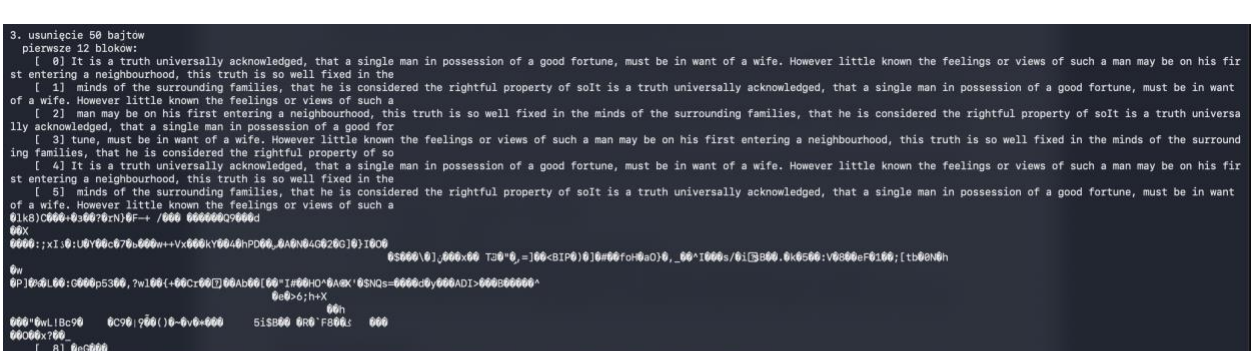
```
3. usunięcie 50 bajtów
  pierwsze 12 bloków:
    [  0] It is a truth universally acknowledged, that a single man in po
    [  1] ssession of a good fortune, must be in want of a wife. However
    [  2] little known the feelings or views of such a man may be on his
    [  3] first entering a neighbourhood, this truth is so well fixed in
    [  4] the minds of the surrounding families, that he is considered th
    [  5] e rightful property of soIt is a truth universally acknowledged
    [  6]  
```

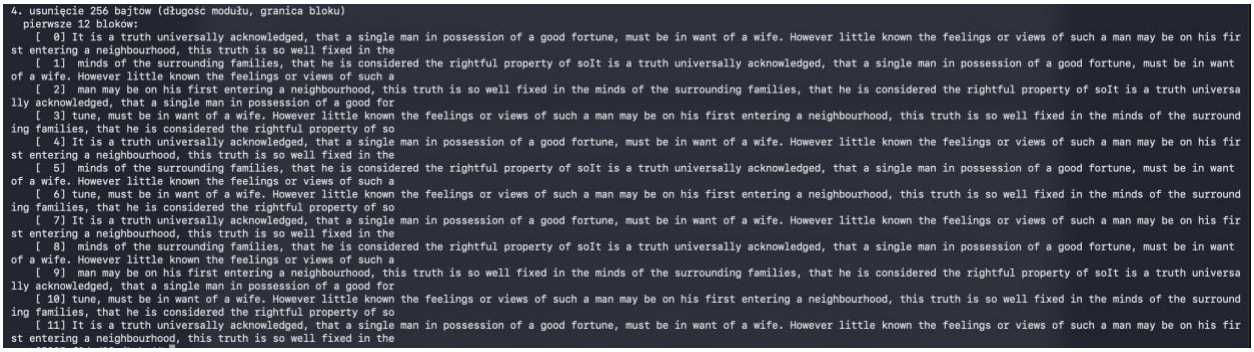
```
4. usunięcie 64 bajtów (długość modułu, granica bloku)
  pierwsze 12 bloków:
    [  0] It is a truth universally acknowledged, that a single man in po
    [  1] ssession of a good fortune, must be in want of a wife. However
    [  2] little known the feelings or views of such a man may be on his
    [  3] first entering a neighbourhood, this truth is so well fixed in
    [  4] the minds of the surrounding families, that he is considered th
    [  5] e rightful property of soIt is a truth universally acknowledged
    [  6]  want of a wife. However little known the feelings or views of
    [  7] such a man may be on his first entering a neighbourhood, this t
    [  8] ruth is so well fixed in the minds of the surrounding families,
    [  9]  that he is considered the rightful property of soIt is a truth
    [ 10]  universally acknowledged, that a single man in possession of a
    [ 11]  good fortune, must be in want of a wife. However little known
```

RSA 2048

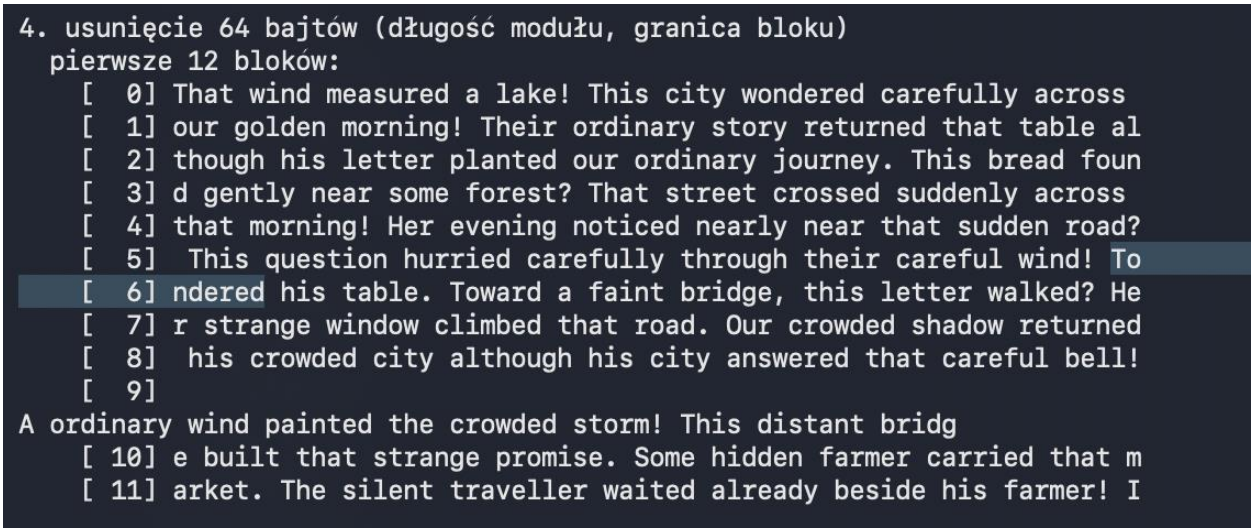

1. zmiana 1 bajtu
pierwsze 12 blokow:
[0] It is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want of a wife. However little known the feelings or views of such a man may be on his fir
st entering a neighbourhood, this truth is so well fixed in the
[1] minds of the surrounding families, that he is considered the rightful property of soIt is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want
of a wife. However little known the feelings or views of such a
[2] man may be on his first entering a neighbourhood, this truth is so well fixed in the minds of the surrounding families, that he is considered the rightful property of soIt is a truth universa
lly acknowledged, that a single man in possession of a good for
[3] tune, must be in want of a wife. However little known the feelings or views of such a man may be on his first entering a neighbourhood, this truth is so well fixed in the minds of the surround
ing families, that he is considered the rightful property of so
[4] It is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want of a wife. However little known the feelings or views of such a man may be on his fir
st entering a neighbourhood, this truth is so well fixed in the
[5] minds of the surrounding families, that he is considered the rightful property of soIt is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want
of a wife. However little known the feelings or views of such a
[6] ZW2u
[7] tune, must be in want of a wife. However little known the feelings or views of such a man may be on his first entering a neighbourhood, this truth is so well fixed in the minds of the surround
ing families, that he is considered the rightful property of so
[8] It is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want of a wife. However little known the feelings or views of such a man may be on his fir
st entering a neighbourhood, this truth is so well fixed in the
[9] minds of the surrounding families, that he is considered the rightful property of soIt is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want
of a wife. However little known the feelings or views of such a
[10] man may be on his first entering a neighbourhood, this truth is so well fixed in the minds of the surrounding families, that he is considered the rightful property of soIt is a truth universa
lly acknowledged, that a single man in possession of a good for
[11] tune, must be in want of a wife. However little known the feelings or views of such a man may be on his first entering a neighbourhood, this truth is so well fixed in the minds of the surround
ing families, that he is considered the rightful property of so


2. usunięcie 12 bajtu
pierwsze 12 blokow:
[0] It is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want of a wife. However little known the feelings or views of such a man may be on his fir
st entering a neighbourhood, this truth is so well fixed in the
[1] minds of the surrounding families, that he is considered the rightful property of soIt is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want
of a wife. However little known the feelings or views of such a
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3. usunięcie 50 bajtow
pierwsze 12 blokow:
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Inny tekst aby było widać usunięcie



1.7

Tak, szczególnie przy tekstach o niskiej entropii zastosowanie większego klucza pozwala na ukrycie części okresowości

1.8

Długość klucza wpływa na okresowość i występowanie pików na wykresach.

1.9

Tak, tekst jawny o niskiej entropii po zaszyfrowaniu również posiada niższą entropię

1.11

Algorytmy symetryczne mają podobne czasy szyfrowania i deszyfrowania, natomiast algorytmy asymetryczny mają znacząco dłuższe czasy deszyfrowania

1.12

Zmiana 1 bajtu powoduje uszkodzenie w jednym bloku

Usunięcie jednego lub kilku bajtów powoduje uszkodzenie pliku od momentu zmiany

Usunięcie fragmentu długości modułu powoduje zniknięcie dokładnie jednego bloku

Skala zależy od miejsca i typu uszkodzenia

1.13

Tak wystarczy usunąć długość bloku

1.14

Zalety asymetrycznych:

- Bezpieczna dystrybucja kluczy
- Jeden klucz do wielu komunikacji

Wady asymetrycznych:

- Wolniejszy szyfrowanie i deszyfrowanie

1.15

Symetryczne do szyfrowania dużych ilości danych

Asymetryczne wymiany kluczy sesyjnych, certyfikaty SSL/TLS i PKI